Separable Differential Equations

Section 1: General Solutions of Separable Equations

1 Solve the differential equation $\frac{dy}{dx} = \frac{2x}{y}$.

a. $y^2 = 2x^2 + C$

b. $y = x^2 + C$

c. $y^2 = x^2 + C$

d. $y = 2x^2 + C$

2 Find the general solution to $\frac{dy}{dx} = 3x^2y$.

a. $y = Ce^{x^3}$

b. $y = e^{x^3} + C$

c. $y = Cx^3$

d. $y = 3x^3 + C$

3 Find the general solution of $\frac{dy}{dx} = (1 + y^2)e^x$.

a. $\arctan(y) = e^x + C$

b. $y = \tan(e^x) + C$

c. $y = e^x + C$

d. $\ln(1 + y^2) = e^x + C$

Section 2: Initial Value Problems (IVPs)

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4 Solve the initial value problem $\frac{dy}{dx} = 2xy$ with $y(0) = 3$.

a. $y = 3e^{x^2}$

b. $y = e^{x^2} + 2$

c. $y = 3e^{2x}$

d. $y = 3x^2 + 3$

5 Solve $\frac{dy}{dx} = -\frac{x}{y}$ with $y(3) = 4$.

a. $x^2 + y^2 = 25$

b. $x^2 - y^2 = -7$

c. $y = -x + 7$

d. $x^2 + y^2 = 7$

6 Solve $\frac{dy}{dt} = ky$ given $y(0) = 50$ and $y(2) = 200$. What is the formula for $y(t)$?

a. $y(t) = 50 \cdot 2^t$

b. $y(t) = 50e^{2t}$

c. $y(t) = 50 \cdot 4^t$

d. $y(t) = 50 + 75t$

Section 3: Slope Fields and Solution Curves

7 Along which curve in the xy-plane do the line segments in the slope field for $\frac{dy}{dx} = x - y$ have slope 0?

a. Along the line $y = x$

b. Along the x-axis ($y = 0$)

c. Along the y-axis ($x = 0$)

d. Along $y = -x$

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8 A slope field shows line segments that depend ONLY on y (horizontal rows of equal slopes). Which differential equation could this represent?

a. $\frac{dy}{dx} = y(2 - y)$

b. $\frac{dy}{dx} = x^2 - y$

c. $\frac{dy}{dx} = xy$

d. $\frac{dy}{dx} = 3x + 1$

9 For $\frac{dy}{dx} = x^2 + y^2$, what can be said about all non-origin slopes in the slope field?

a. All slopes are strictly positive

b. Slopes are positive in quadrants I and III only

c. Slopes are 0 everywhere

d. Slopes are negative when $x < 0$

Section 4: Logistic Growth Models

10 In the logistic differential equation $\frac{dP}{dt} = 0.05P\left(1 - \frac{P}{800}\right)$, what is the carrying capacity M?

a. 800

b. 0.05

c. 400

d. 16000

11 At what population P is the growth rate $\frac{dP}{dt}$ maximized in the logistic equation $\frac{dP}{dt} = 0.05P\left(1 - \frac{P}{800}\right)$?

a. $P = 400$

b. $P = 800$

c. $P = 200$

d. $P = 0$

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12 What is $\lim_{t\to\infty} P(t)$ for any solution to $\frac{dP}{dt} = 0.05P\left(1 - \frac{P}{800}\right)$ with $P(0) = 50$?

a. 800

b. ∞

c. 400

d. 50

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